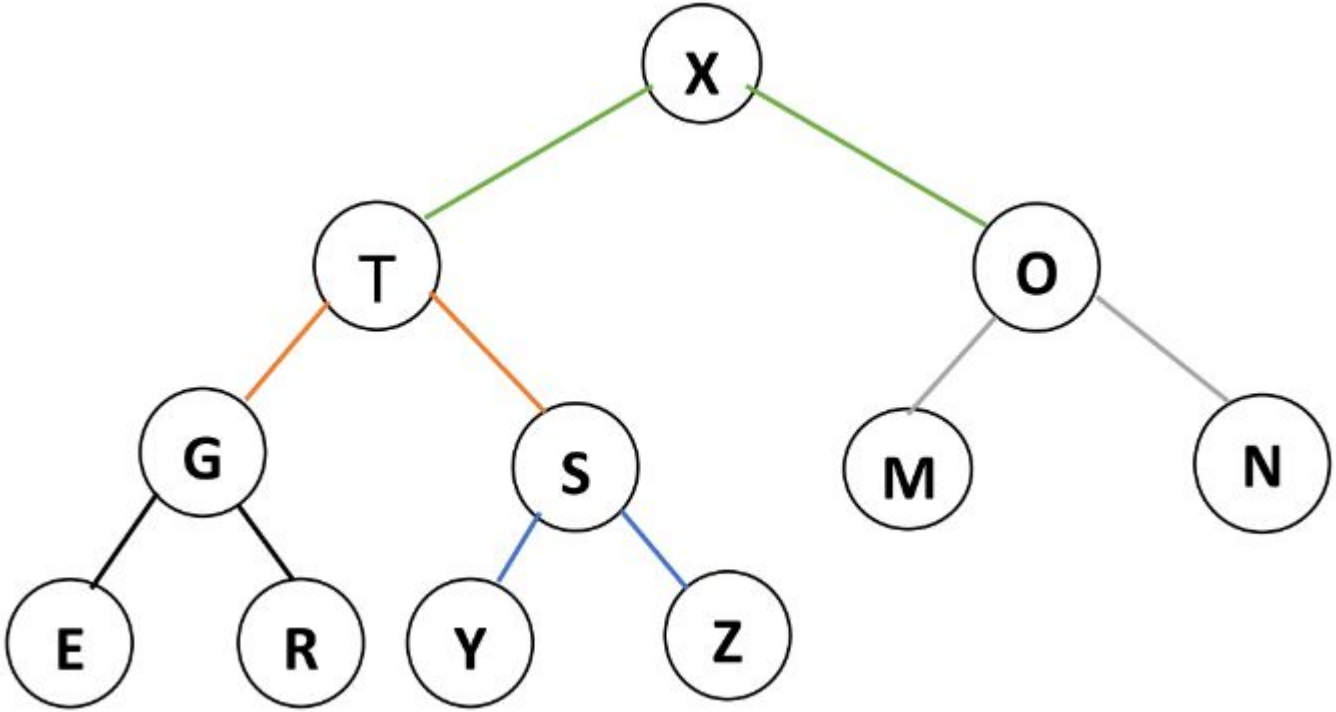
Data Structures

Heaps

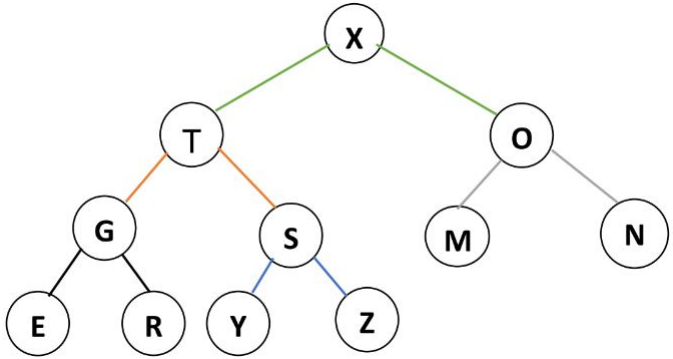
Slides taken from Prantik Paul [PKP]

Heap - Special Binary Tree



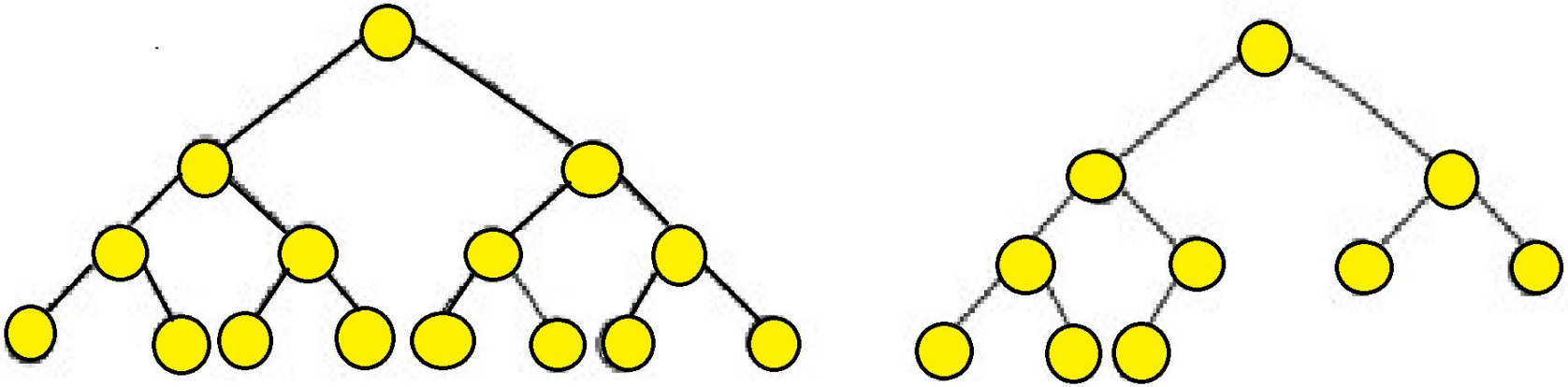
Heap - Properties

- Complete Binary Tree
- Satisfy Heap Property



- Value of Parent >= Value of Children MaxHeap
- Value of Parent <= Value of Children MinHeap

Binary Tree - Complete Binary Tree

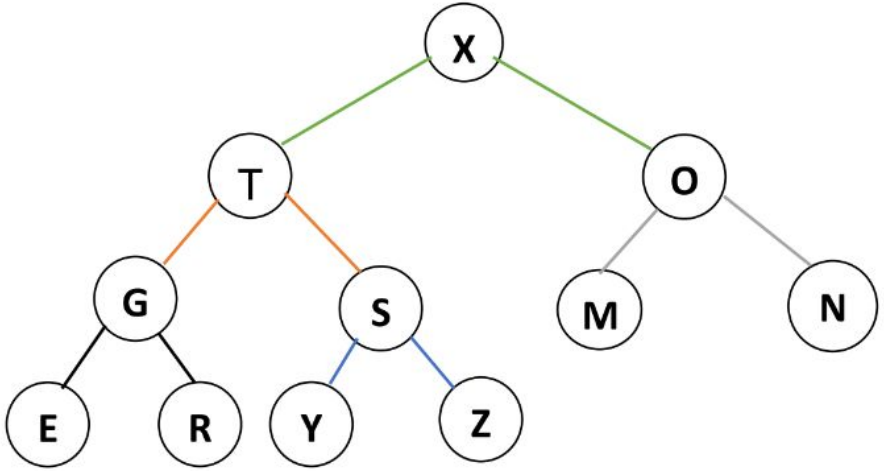


All levels filled starting from LEFT

Heap Simulation

https://visualgo.net/en/heap?slide=1

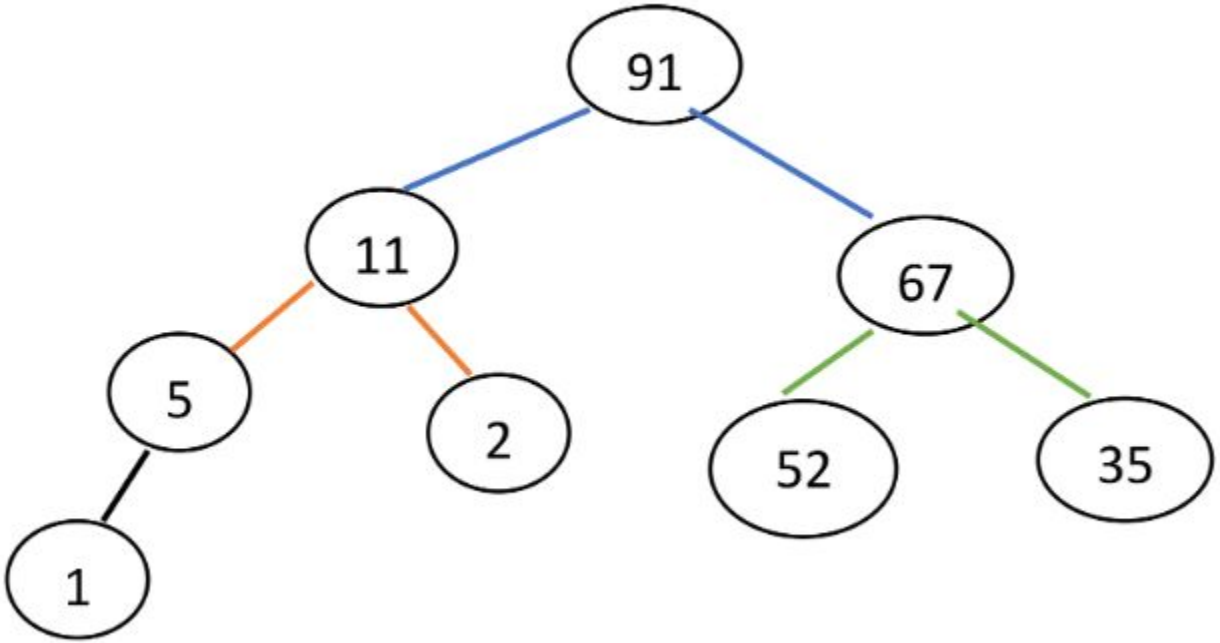
Heap - Representation



1	2	3	4	5	6	7	8	9	10	11	12	
X	T	O	G	S	M	N	E	R	Y	Z	C	

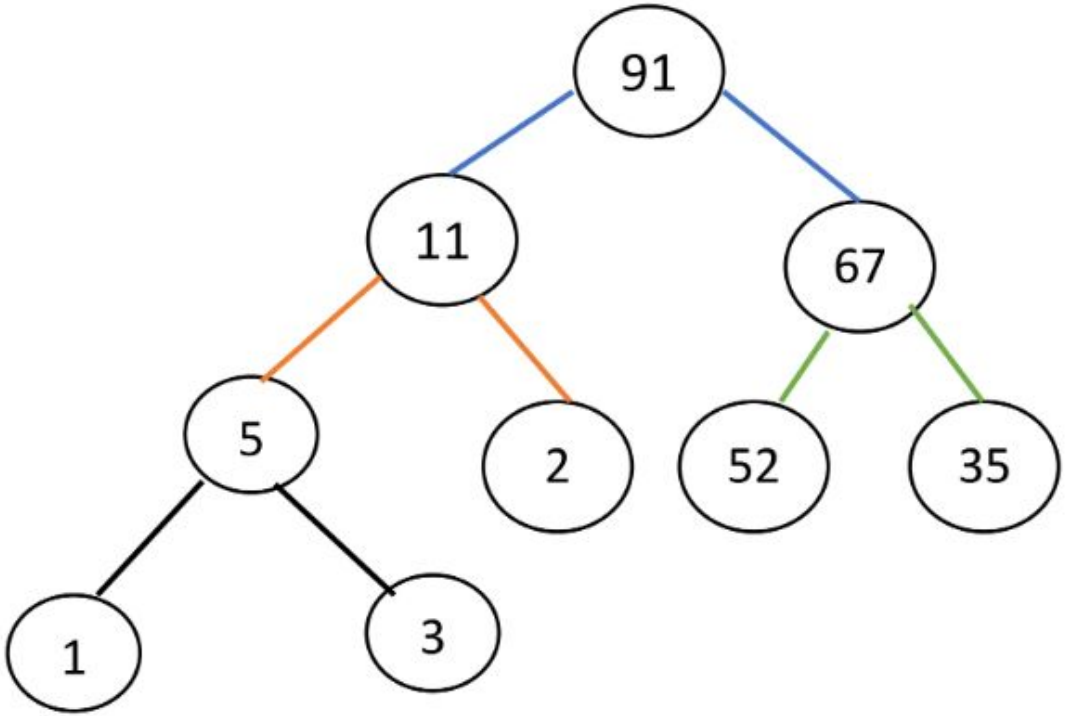
Heap - Insert

Insert 3



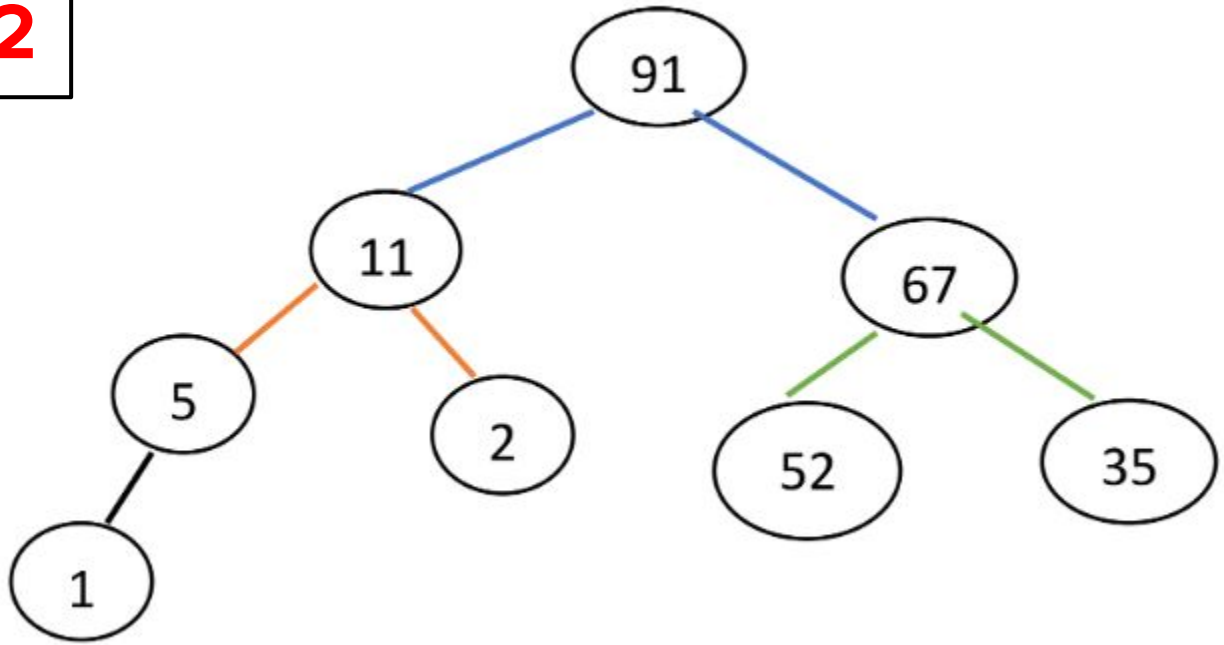
Heap - Insert

Insert 3



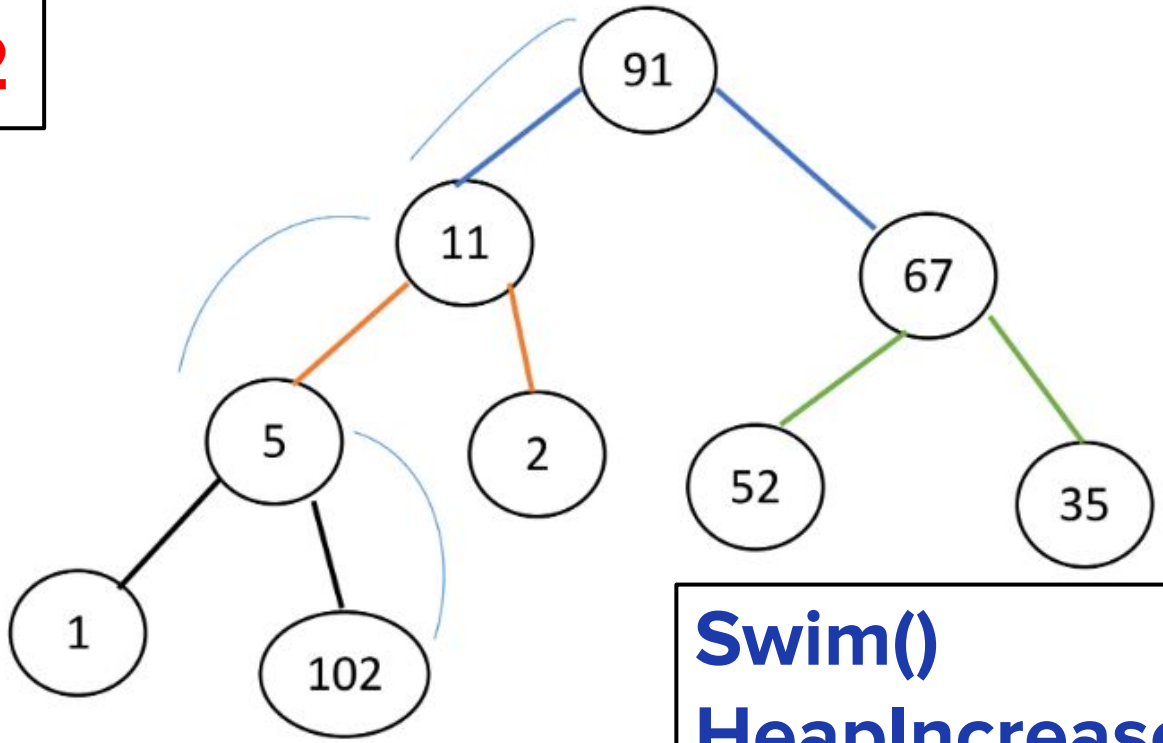
Heap - Insert

Insert 102



Heap - Insert

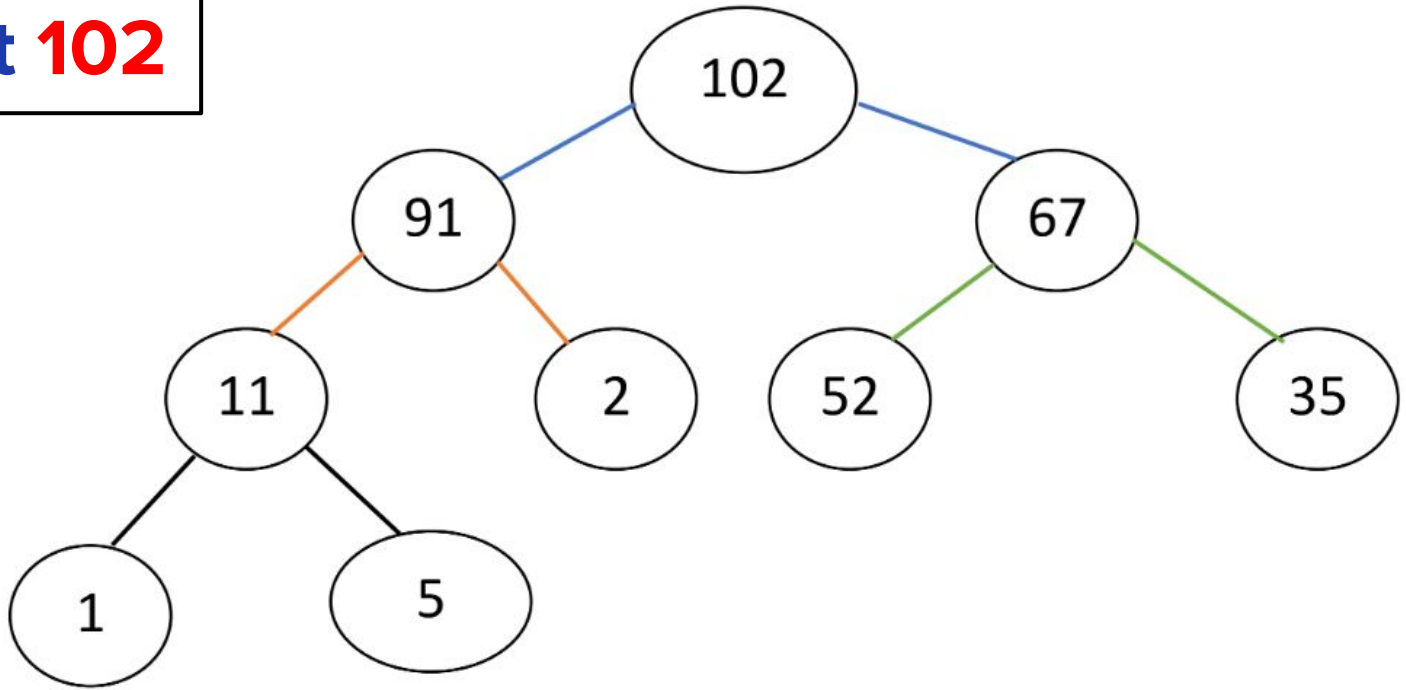
Insert 102



Swim()
HeapIncreaseKey()

Heap - Insert

Insert 102



Heap - Insert (Pseudocode)

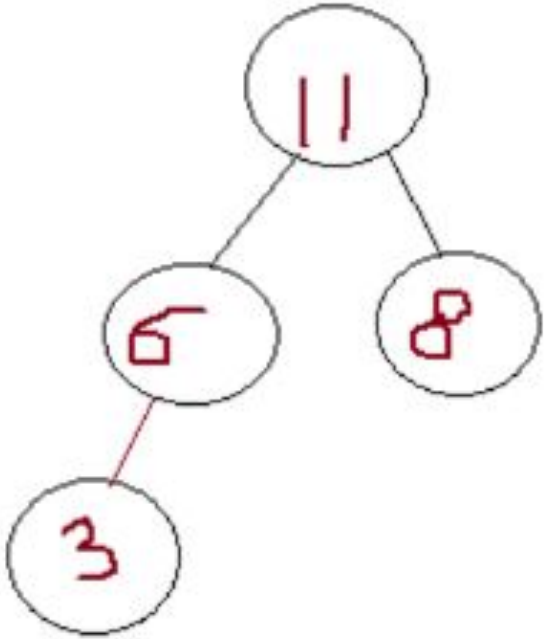
```
insert (H, key){
        size(H) = size(H) + 1;
        H[size] = key;
        swim (H, size);
}
```

Heap - Swim (Pseudocode)

```
swim(H, index){
        if (index <= 1){
                return;
        }else{
                parent = H[index/2];
                if (parent > H[index]){
                        return;
                }else{
                        exchange parent with H[index]
                        swim(H, parent);
                }
}
```

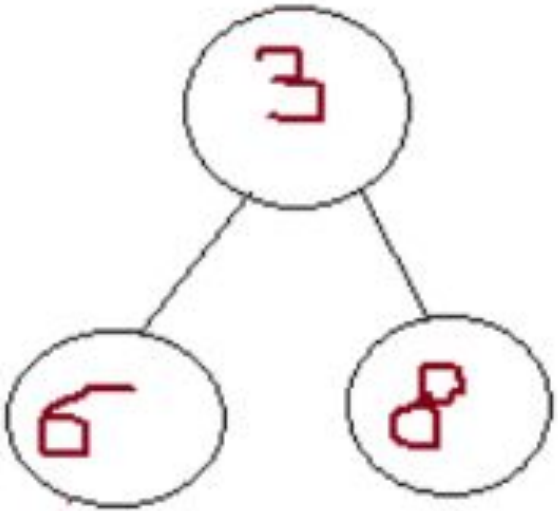
Heap - Delete

Delete Root



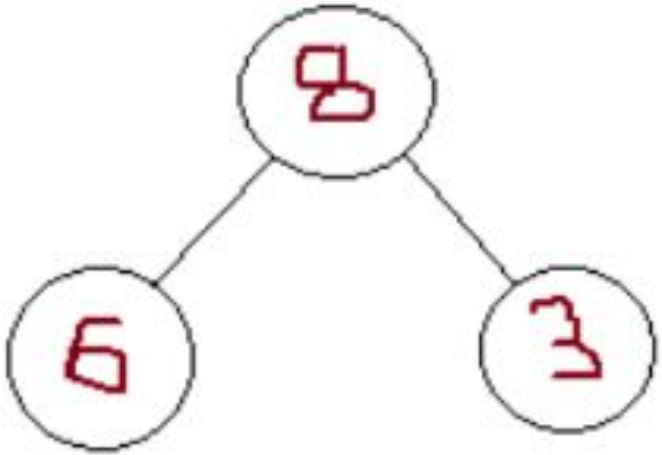
Heap - Delete

Delete Root



Heap - Delete

Delete Root



Sink()
MaxHeapify()

Heap - Delete (Pseudocode)

```
delete(H){
        if (size(H)==0){
                return;
        }else{
                exchange H[1] with H[size]
                size --;
                maxHeapify(H, 1)
        }
}
```

Heap - Sink (Pseudocode)

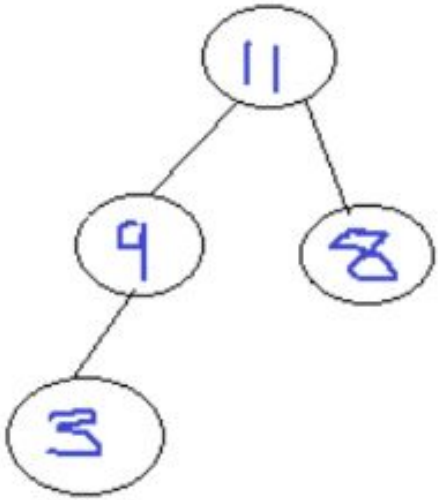
```
maxHeapify(H, index){
        if (size(H) ==0){
                return;
        }else{
                left = 2*index;
                right=2*index+1;
                if (left <= size && right<=size){
                        exchange H[1] with Max (H[left], H[right]);
                        maxHeapify(Max (left, right));
                }else{
                        if (left<= size && right>size){
                        exchange H[1] with (H[left]);
                    }

            }
}
```

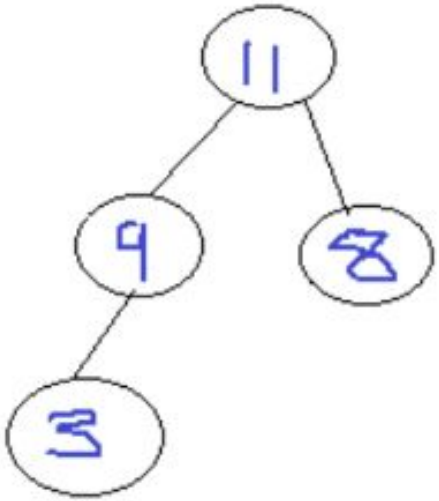
Heap Sort

Delete All Nodes

Each time the max element is deleted



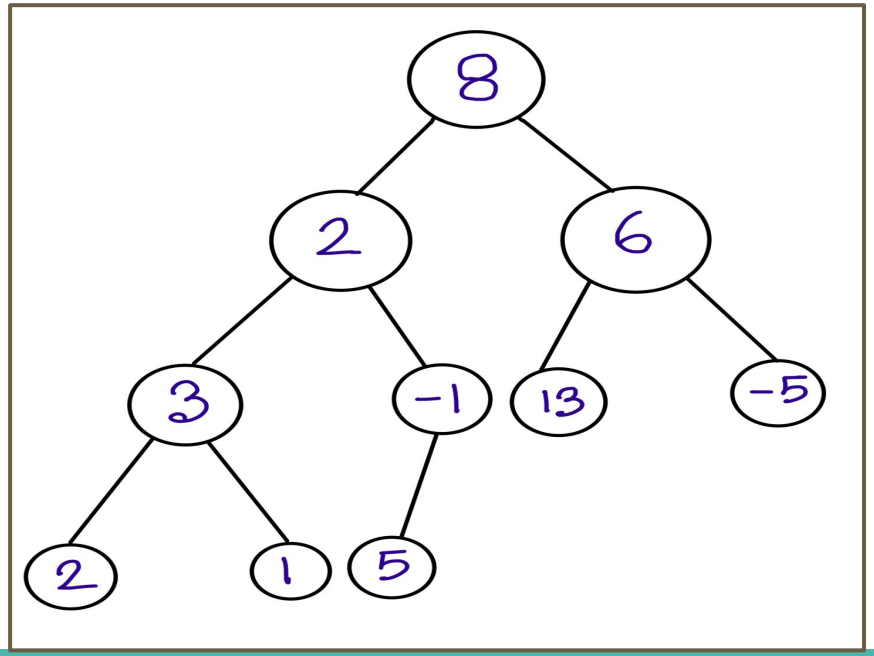
Heap Sort



Heap - Build Heap

N	8	2	6	3	-1	13	-5	2	1	5

Tree

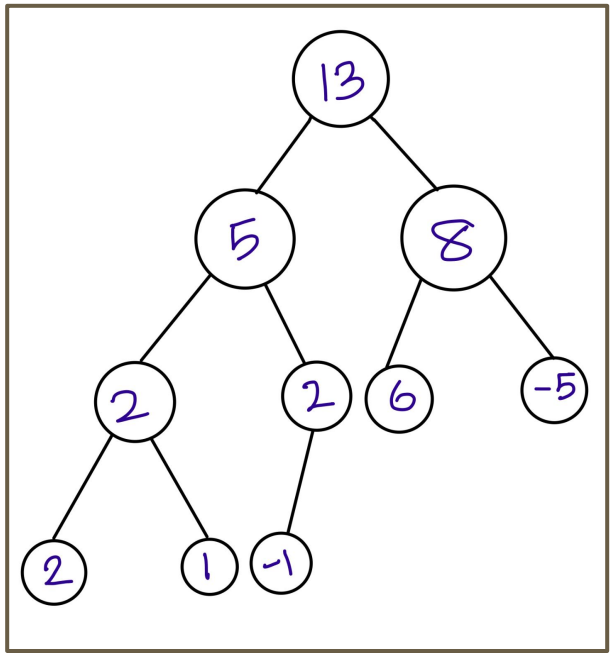


Heap - Build Heap

8	2	6	3	-1	13	-5	2	1	5

```
for all nodes i = 1 to n{
        swim (H, i);
}
```

Heap

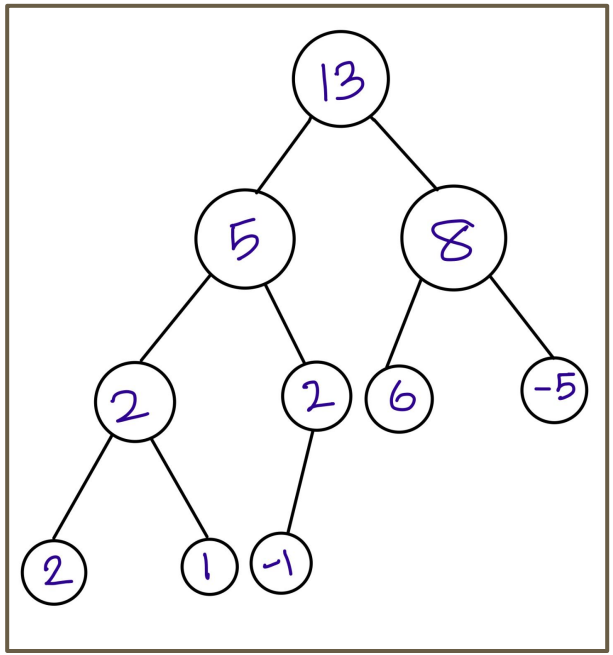


Heap - Build Heap

13	5	8	2	2	6	-5	2	1	-1

```
for all nodes i = 1 to n{
        swim (H, i);
}
```

Heap



Learn More About Heaps

https://www.geeksforgeeks.org/heap-data-structure/